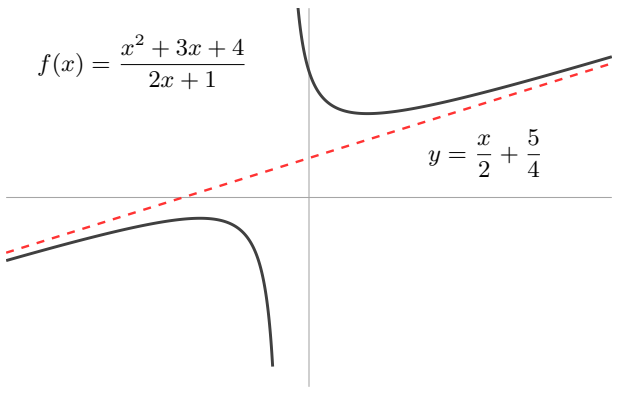


$$f(x) = \frac{x^2 + 3x + 4}{2x + 1}$$

$$y = \frac{x}{2} + \frac{5}{4}$$


The graph shows a rational function  $f(x) = \frac{x^2 + 3x + 4}{2x + 1}$  plotted on a Cartesian coordinate system. The function is represented by a solid black curve with two branches. A dashed red line represents the slant asymptote  $y = \frac{x}{2} + \frac{5}{4}$ . The vertical asymptote is the line  $x = -\frac{1}{2}$ , which is not explicitly drawn but is implied by the behavior of the function. The curve approaches the vertical asymptote as  $x \rightarrow -\frac{1}{2}^-$  and  $x \rightarrow -\frac{1}{2}^+$ . The curve also approaches the slant asymptote as  $x \rightarrow \pm\infty$ .